

LESSON 6-7 • SESSION 1 • PRACTICE • FORM A

Factors and Greatest Common Factor

Learning target – I can list the factors of two numbers and find their greatest common factor. · 6.NS.B.4

REMEMBER

- A **factor** divides a number with no remainder. A **common factor** divides both numbers.
- The **GCF** is the greatest factor two numbers share. Check it by dividing it into both.
- **Prime factorization**: split until every factor is prime:
 $63 = 7 \times 3 \times 3$.

WORKED EXAMPLE

Find the GCF of 20 and 30.

Step 1 Factors of 20: 1, 2, 4, 5, 10, 20.**Step 2** Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30.**Step 3** Shared factors: 1, 2, 5, 10. The greatest is 10.**Answer** **GCF = 10****WORDS** **factor** (*factor*) · **greatest common factor** (*máximo común divisor*) · **prime factorization** (*factorización prima*)**PRACTICE***show your work · label units*

1. Find the GCF of each pair.

a. 16 and 24 _____**b.** 18 and 30 _____**c.** 21 and 35 _____**d.** 15 and 25 _____

2. **Factor Pairs.** Work with the number 45.

Hint Start with 1×45 , then try 2, 3, 4, 5, ...**a.** List every factor pair of 45. _____**b.** List all factors of 45 from least to greatest. _____

3. **Prime Factor Tree.** Build a factor tree for 42 in the box.

Hint Split 42 into 6×7 , then split 6.**a.** Prime factorization of 42: _____**b.** Is 7 a factor of 42? _____*Show your work.*

4. **Gift Bags.** A volunteer has 18 erasers and 27 stickers. Every bag must hold the same number of each, with nothing left over.

a. Greatest number of bags: _____**b.** In one bag: erasers ____ stickers ____*Show your work.*

5. **Pilar's Answer.** Pilar says the GCF of 15 and 25 is 3. Is she right? Give the correct GCF.

Hint Does 3 divide 25 with no remainder?

Show your work.

GCF _____

6. **Zola's Answer.** Zola says the GCF of 14 and 49 is 2. Is she right? Give the correct GCF.

Hint List the factors of 49.

Show your work.

GCF _____

7. **Garden Rows.** A gardener has 28 carrot seeds and 40 bean seeds. She plants identical rows with none left over.

- a. Greatest number of rows: _____
b. In one row: carrot seeds ____ bean seeds ____

Show your work.

8. **Party Bags.** A host has 48 stickers and 60 erasers for identical bags with none left over.

- a. Find the greatest number of bags. _____
b. How many of each item go in one bag? _____
c. Explain why 10 bags would not work. _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Farrah's Work.** Farrah said the GCF of 9 and 30 is 9, because 9 is a factor of 9. Explain her mistake. Then find the correct GCF.

TEST PRACTICE

choose the best answer

10. The GCF of 12 and 18 is –

A 3 **B** 6 **C** 36 **D** 2

11. A factor of a number divides it with –

A a remainder of 1 **B** a decimal answer **C** no remainder **D** a remainder of 2

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

9 A	6 X	12 R	3 Q	5 C	2 × 3 × 7 F	2 Z	7 T	4 O
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#3a	#4a	#5	#6	#7a	#8a
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____